

5/22/2026

WEEK 1: Technical Support Fundamentals and Networking

Smarties Group Members

Siphelele Malotana

Nobahle Nzimande

Sipe Nonyukela

Liam Alexander

Table of Contents

Task 1: Business Requirements Analysis	2
Task 2: Hardware Inventory	7
Task 3: Software Inventory	9
Task 4: Initial Network Diagram.....	15
Task 5: Troubleshooting Scenarios.....	17
Scenario 1 — Desktop Not Powering On.....	17
Scenario 2 — No Internet Connectivity.....	18
Scenario 3 — Printer Offline	19

Task 1: Business Requirements Analysis

1. Company Overview & Organisational Structure

Ubuntu Innovations (Pty) Ltd is a rapidly growing technology startup based in Cape Town, South Africa. The company is relocating to a new office and requires a fully documented IT infrastructure built from scratch across its 25-person workforce.

Organisational Structure	Company Context
Department	Industry: Information Technology / Software Development
Executive Management	
Finance	Stage: Early-growth startup
Human Resources	Location: Cape Town, Western Cape, South Africa
Sales & Marketing	Office: New premises — greenfield IT deployment
Software Development	
TOTAL: 25 employees	IT Maturity: Low — no existing legacy infrastructure
	Operating Hours: Mon–Fri, 08:00–17:00 SAST
	Users: 25 full-time employees across 5 departments

2. Business Requirements Summary

The following requirements were identified across all departments and must be addressed through the IT infrastructure design, security policies, and support procedures.

#	Category	Requirement	Priority
1	Connectivity	Wired & wireless LAN for all 25 users; business-grade firewall router	Critical
2	User Management	Individual accounts with dept-based security groups & RBAC	Critical

3	File Storage	NAS with departmental folder structure and access permissions	Critical
4	Printing	Two shared network laser printers accessible across the LAN	Medium
5	Data Security	Antivirus, device encryption (laptops), MFA, patch management	Critical
6	Backup & Recovery	Daily incremental, weekly full, monthly archive; cloud offsite copy	High
7	Applications	Microsoft 365 for all staff; dept-specific software per section 3	Critical
8	Power Protection	Two 1500VA UPS units protecting core networking equipment	High
9	Documentation	Troubleshooting guide and IT support procedures for staff	Medium

3. Business Application Requirements by Department

Department	Required Software	Notes
All Staff (25)	Microsoft 365, Antivirus (Bitdefender), Zoom, Web Browser	Core productivity and security baseline for every device
Executive Mgmt (3)	Power BI, Asana, BitLocker encryption	Laptops must be fully encrypted; mobility-focused
Finance (4)	Sage Business Cloud, Advanced Excel, Seclore DRM	SARS-compliant; payroll data requires strict access control
Human Resources (3)	SimplePay, SharePoint (M365), DocuSign	Employee records need privacy controls and digital signing
Sales & Marketing (7)	HubSpot CRM, Canva Pro, Hootsuite, Google Analytics	Cloud-dependent; stable internet access is critical
Software Dev (8)	VS Code, Git, GitHub, Docker, Ubuntu Linux 22.04	Developer workstations run Linux; elevated permissions needed

4. Operational Challenges

Six key operational challenges have been identified that the IT infrastructure design must address from the outset.

Challenge	Description	Challenge	Description
Greenfield Deployment	All infrastructure must be built from scratch before operations begin.	Data Segregation	Finance and HR hold sensitive data requiring strict access controls.
Load Shedding	SA power outages risk downtime and data loss without UPS protection.	Cybersecurity Threats	Phishing and ransomware are high-probability risks requiring MFA and AV.
Scalability	Infrastructure must accommodate projected staff growth without a rebuild.	Software Licensing	25 devices require tracked, compliant licences across all applications.

5. Data Security & Backup Requirements

Data Security Controls	Backup & Recovery Schedule		
Antivirus & endpoint protection on all 25 devices	Frequency	Type	Retention
BitLocker encryption on all 5 executive laptops	Daily	Incremental	7 days rolling
Multi-factor authentication (MFA) for all accounts	Weekly	Full backup	4 weeks
Security patching within 72hrs (critical) / 14 days (standard)	Monthly	Archive	12 months
Written Acceptable Use Policy issued to all staff	Continuous	Cloud offsite	Real-time
Physical access controls for server room and network rack	RTO: < 4 hours RPO: < 24 hours		
	NAS with RAID-5 as primary backup target		

6. Connectivity & Network Requirements

Wired & Wireless Network	IP Addressing Plan (Preview)	
Gigabit Ethernet for all 25 desktop workstations	Device Type	Subnet
24-port managed switch for internal LAN	Servers / NAS	192.168.10.0/24
2× Wi-Fi 6 dual-band access points for wireless coverage	User Devices	192.168.20.0/24
Business-grade firewall router (NAT, VPN, traffic filtering)	Printers	192.168.30.0/24
VLAN segmentation: servers, users, printers, Wi-Fi	Wi-Fi Clients	192.168.40.0/24
2× 1500VA UPS units for core equipment power protection		

7. Key Recommendations

#	Recommendation
1	Deploy a VLAN-segmented network from day one to ensure security and scalability as the company grows.
2	Implement centralised identity management (Active Directory or equivalent) for user accounts and group policy enforcement.
3	Enforce a zero-trust security baseline from the outset: MFA on all accounts, endpoint AV, encrypted laptops, and a signed AUP.
4	Select all hardware with a minimum 3-year South African vendor warranty to reduce support risk and downtime.
5	Design the backup strategy to satisfy both on-premises (NAS/RAID-5) and offsite (cloud replication) requirements.
6	Document all configurations, procedures, and troubleshooting guides immediately to reduce single-point-of-knowledge risk.
7	Plan a 30% growth buffer in IP addressing, switch port capacity, and NAS storage to accommodate future hiring.

Task 2: Hardware Inventory

Hardware Inventory Table

The table below details all recommended hardware assets for Ubuntu Innovations (Pty) Ltd. All costs are estimated in South African Rand (ZAR).

Asset	Qty	Suggested Specification	Unit Cost	Total Cost	Business Justification	Warranty
Desktop Computers	20	Intel Core i5, 16GB RAM, 512GB SSD, Windows 11 Pro	R12,000	R240,000	General office use across all 5 departments	3-year on-site warranty recommended
Laptops	5	Intel Core i7, 16GB RAM, 512GB SSD, Windows 11 Pro + BitLocker	R18,000	R90,000	Mobility for Executive Management team	3-year on-site warranty recommended
Router / Firewall	1	Business-grade firewall router (e.g. Cisco RV340 / Fortinet 40F)	R8,500	R8,500	Secure internet access, NAT, VPN and traffic filtering	3-year vendor support contract
Managed Switch	1	24-port Gigabit managed switch (e.g. Cisco SG350-28)	R4,500	R4,500	Internal LAN for all 25 wired devices plus spare ports	3-year vendor support contract
Wireless Access Points	2	Dual-band Wi-Fi 6 AP (e.g. Ubiquiti UniFi U6-LR)	R3,200	R6,400	Full wireless coverage across the office floor	2-year manufacturer warranty
Network Printers	2	Monochrome laser printer with network card (e.g. HP)	R5,500	R11,000	Shared departmental printing - low cost per page	1-year warranty; service

		LaserJet Pro M404dn)				contract optional
NAS Device	1	8TB RAID-5 NAS (e.g. Synology DS423+ with 4x4TB WD Red drives)	R14,000	R14,000	Centralised file storage and daily backup target	3-year manufacturer warranty
UPS Units	2	1500VA UPS (e.g. APC Smart-UPS 1500VA LCD)	R3,800	R7,600	Power protection for network rack during load shedding	2-year manufacturer warranty

TOTAL ESTIMATED BUDGET	R382,000
---------------------------------------	-----------------

Notes & Recommendations

- All unit costs are estimates in South African Rand (ZAR) and may vary by supplier and market conditions at time of purchase.
- Desktop and laptop units must each include a valid Windows 11 Pro licence.
- The NAS device must be configured in RAID-5 to provide single-disk fault tolerance.
- UPS units protect the core network rack (router, switch, NAS) against South African load shedding.
- A hardware refresh cycle of 3-5 years is recommended for desktops and laptops.
- The 24-port switch provides spare ports to accommodate future staff additions.

Task 3: Software Inventory

Document Overview

This software inventory documents all recommended software for Ubuntu Innovations (Pty) Ltd, a 25-employee technology startup relocating to a new office in Cape Town. Software is organised by category: company-wide tools and department-specific applications. Licensing types, cost models, and deployment notes are included.

1. Company-Wide Software

The following software is required across all 25 users regardless of department.

Software	Purpose	Licence Type	Cost Model	Deployment
Ubuntu 22.04 LTS	Desktop operating system	Open Source (GPL)	Free	All desktops & laptops
LibreOffice 7.x	Word processing, spreadsheets, presentations	Open Source (MPL)	Free	All workstations
Google Workspace (Business Starter)	Email, Calendar, Drive, Docs, Meet	SaaS - Per user	~R111/user/month (~R2,775/mo)	Web & desktop app
Bitwarden (Teams)	Password manager	SaaS - Per user	~R55/user/month (~R1,387/mo)	Browser extension + app
ClamAV + Malwarebytes for Teams	Antivirus & endpoint protection	Open Source / Commercial	Free / ~R74/user/month	All endpoints
UFW (Uncomplicated Firewall)	Host-based firewall	Open Source	Free	All Linux machines
Timeshift	Local system snapshots	Open Source (GPL)	Free	All workstations

CUPS (Print Server)	Shared printer management	Open Source	Free	Server / all users
OpenVPN / WireGuard	Remote access VPN	Open Source	Free	Server + clients
Remmina	Remote desktop client	Open Source	Free	IT admin machines

2. Department-Specific Software

2.1 Executive Management (3 Users) — Strategic planning, communication, and reporting tools

Software	Purpose	Licence	Notes
Power BI / Google Looker Studio	Business dashboards & KPI reporting	Free / Paid	Looker Studio free; Power BI Pro ~R185/user
Slack (Pro)	Internal team messaging	SaaS - Per user	Optional; Google Chat already in Workspace
DocuSign / Xodo Sign	Electronic document signing	SaaS	~R277/user/month
Trello / Asana (Free tier)	Project tracking	Freemium SaaS	Free for small teams

2.2 Finance Department (4 Users) — Accounting, payroll, and compliance software

Software	Purpose	Licence	Notes
Sage Business Cloud Accounting	Accounting & invoicing (SA-compliant)	SaaS - Per user	~R350/user/month; VAT/SARS ready
Microsoft Excel (via M365 Basic)	Financial modelling & spreadsheets	SaaS - Per user	~R111/user/month; alternative to LibreOffice

PaySpace / SimplePay	Payroll & HR (SA legislation compliance)	SaaS	PaySpace from ~R200/month
KeePassXC	Secure credential storage for finance systems	Open Source	Free

2.3 Human Resources (3 Users) — Employee management, onboarding, and leave tracking

Software	Purpose	Licence	Notes
BambooHR / Zoho People	HR management, leave & performance tracking	SaaS - Per user	Zoho from ~\$1.25/user; BambooHR from \$6/user
Google Forms	Employee surveys & onboarding forms	Included in Workspace	Free
Canva (Pro)	HR communications, onboarding materials	SaaS	~R240/month/team
Zoom / Google Meet	Interview & HR meetings	Included in Workspace	Free

2.4 Sales and Marketing (7 Users) — CRM, campaign management, and content creation

Software	Purpose	Licence	Notes
HubSpot CRM (Free tier)	Customer relationship management	Freemium SaaS	Free up to 1M contacts; Paid from ~R832/month
Mailchimp / Brevo	Email marketing campaigns	Freemium SaaS	Free up to 500 contacts (Brevo)
Canva (Pro)	Marketing graphics & social media posts	SaaS	~R240/month team
Google Analytics	Website traffic analysis	Free (SaaS)	Free
Meta Business Suite / Buffer	Social media scheduling & management	Freemium SaaS	Buffer free for 3 channels

2.5 Software Development (8 Users) — Development environments, version control, collaboration

Software	Purpose	Licence	Notes
Visual Studio Code	Code editor / IDE	Open Source (MIT)	Free
Git + GitHub / GitLab	Version control & code repository	Open Source / SaaS	GitHub Free tier available
Docker Desktop (Linux)	Containerisation & dev environments	Open Source (Apache 2)	Free on Linux
Postman	API testing & documentation	Freemium SaaS	Free tier available
VirtualBox	Local VM testing environments	Open Source (GPL)	Free
Jira / GitHub Issues	Sprint & issue tracking	Freemium SaaS	Free up to 10 users (Jira)
DBeaver Community	Database management & SQL editor	Open Source	Free

3. Server and IT Infrastructure Software

Software deployed on the company server and NAS to support network services.

Software	Purpose	Licence	Notes
Ubuntu Server 22.04 LTS	NAS / file server OS	Open Source	Free; LTS until 2027
Samba	Windows-compatible file sharing (SMB/CIFS)	Open Source (GPL)	Free
OpenLDAP	Centralised user directory services	Open Source	Free
Netdata / Zabbix	Network & server monitoring	Open Source	Free

Rsync + cron	Automated backup scripting	Open Source	Free
Veeam Agent for Linux (Free)	Backup and recovery	Freeware	Free for 1 server
Bind9 / dnsmasq	Internal DNS resolution	Open Source	Free
ISC DHCP Server	Dynamic IP assignment	Open Source	Free

4. Software Update Procedures

Update Type	Frequency	Method	Responsible
OS & package updates (Ubuntu)	Weekly (automated) + manual monthly review	apt update & apt upgrade or unattended-upgrades	IT Administrator
Antivirus definitions	Daily (automated)	ClamAV freshclam daemon; Malwarebytes auto-update	IT Administrator
SaaS applications	Automatic (vendor-managed)	Browser / app auto-update	Vendors / IT to monitor
Open-source desktop apps	Monthly patch cycle	apt or Snap package manager	IT Administrator
Security patches (critical)	Within 48 hours of release	Emergency apt patch + restart where required	IT Administrator

5. Estimated Monthly Software Licensing Cost Summary

Category	Solution	Est. Monthly Cost (ZAR)
----------	----------	-------------------------

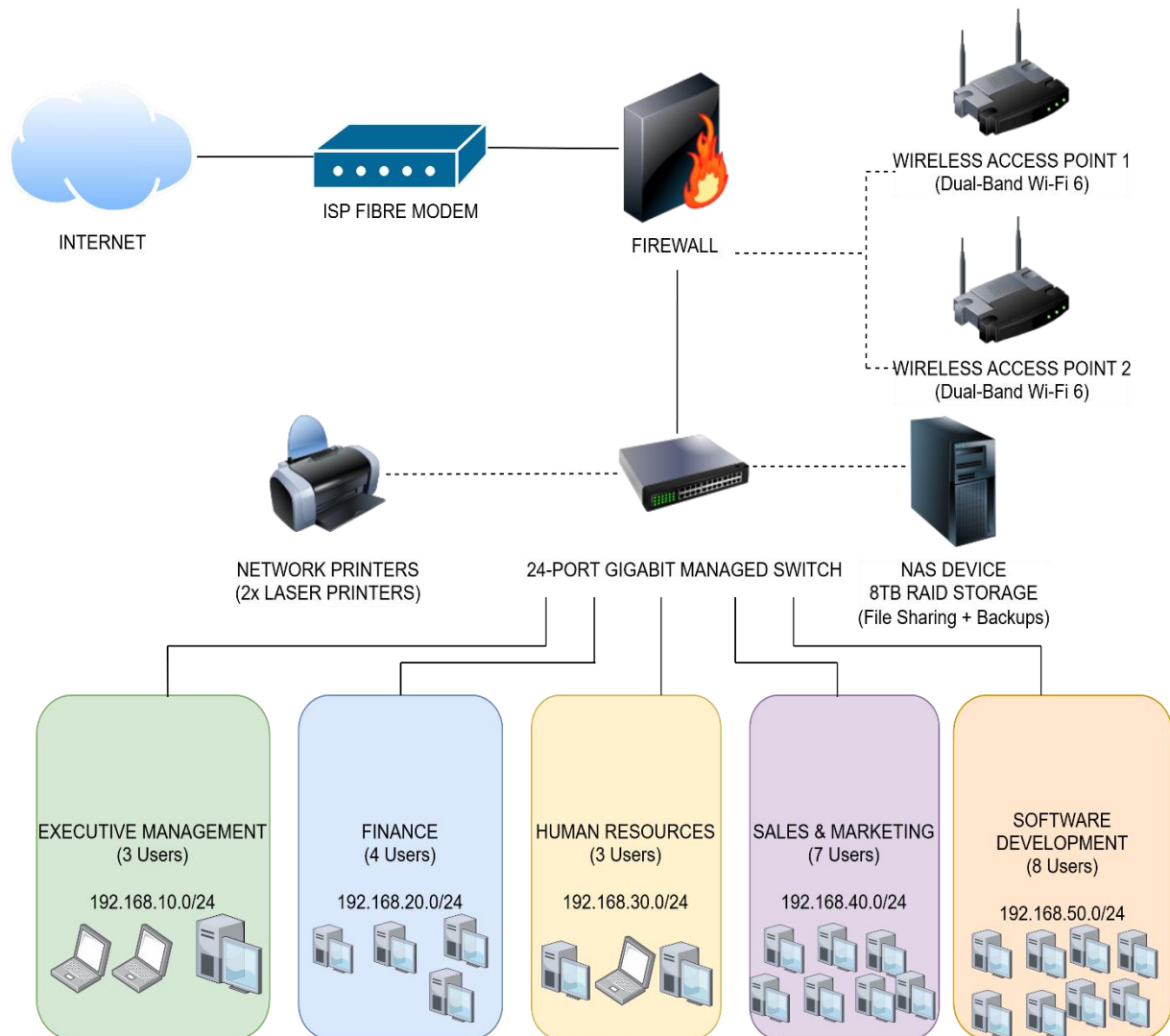
Productivity Suite	Google Workspace Business Starter (25 users)	R2,775
Password Manager	Bitwarden Teams (25 users)	R1,387
Antivirus (Endpoint)	Malwarebytes Teams (25 users)	R1,850
Electronic Signing	DocuSign / Xodo (3 exec users)	R832
Accounting	Sage Business Cloud (4 finance users)	~R1,110
Payroll	SimplePay / PaySpace	~R462
HR Management	Zoho People (3 HR users)	~R74
CRM	HubSpot CRM Free	\$0
Marketing	Canva Pro (team)	R240
Developer Tools	GitHub Free / Jira Free	\$0
Open Source Tools	All other software listed above	\$0

ESTIMATED TOTAL	* Open-source software constitutes the majority of the stack, significantly reducing licensing expenditure.	~R7472 USD/month
------------------------	--	-------------------------

* Costs are approximate ZAR estimates at ~R18.50/USD. South African Rand equivalent will vary with exchange rates. All open-source software listed carries no licensing cost.

Task 4: Initial Network Diagram

Ubuntu Innovation Network Diagram



Task 5: Troubleshooting Scenarios

Scenario 1 — Desktop Not Powering On

Ticket Information

Issue Title: Desktop Computer Not Powering On

Priority: High

Category: Hardware Support

Reported By: End User

Status: Resolved

Problem Description

The user reported that the desktop computer would not power on after pressing the power button. The system showed no signs of activity and could not be used for work purposes.

Symptoms

- Power button not responding
- No lights displayed on the desktop
- No fan movement or startup sound
- Monitor remained blank
- Keyboard and mouse lights inactive

Possible Causes

- Faulty power cable
- Power supply unit (PSU) failure
- Loose internal power connections
- Damaged wall socket or power outlet
- Faulty motherboard or RAM issue

Troubleshooting Steps

1. Verified that the wall socket was functioning correctly by testing another device.
2. Checked the desktop power cable for visible damage and ensured it was securely connected.
3. Confirmed that the power supply switch at the back of the desktop was turned on.
4. Removed and reconnected the RAM modules to ensure proper seating.

5. Opened the desktop case and checked for loose motherboard power connections.
6. Tested the desktop using a different power cable and power outlet.
7. Performed a power supply test using a spare PSU.

Resolution

The issue was caused by a faulty power supply unit. The PSU was replaced with a functioning unit, and the desktop powered on successfully afterward. The user confirmed that the system was operating normally.

Scenario 2 — No Internet Connectivity

Ticket Information

Issue Title: No Internet Connectivity on Workstation

Priority: Medium

Category: Network Support

Reported By: End User

Status: Resolved

Problem Description

The user reported being unable to access the internet on their workstation. Websites could not load, and online applications were unavailable.

Symptoms

- “No Internet Access” message displayed
- Browser pages failed to load
- Network icon showed limited connectivity
- Unable to ping external websites
- Email and cloud services inaccessible

Possible Causes

- Loose or disconnected network cable
- Router or modem failure
- Incorrect IP configuration
- Disabled network adapter
- ISP service interruption

Troubleshooting Steps

1. Checked whether other devices could access the internet.
2. Verified that the Ethernet cable was properly connected to the workstation and router.
3. Restarted the router and modem.
4. Disabled and re-enabled the network adapter.
5. Ran Windows Network Troubleshooter.
6. Opened Command Prompt and executed:
 - ipconfig /release
 - ipconfig /renew
 - ipconfig /flushdns
7. Tested internet connectivity using the ping google.com command.
8. Updated the network adapter drivers.

Resolution

The issue was caused by an incorrect IP address assignment. Renewing the IP configuration restored internet connectivity successfully. The user confirmed that all online services were functioning properly.

Scenario 3 — Printer Offline

Ticket Information

Issue Title: Network Printer Showing Offline

Priority: Medium

Category: Printer Support

Reported By: Office Staff

Status: Resolved

Problem Description

The office printer appeared offline, preventing users from printing documents from their computers.

Symptoms

- Printer status displayed as “Offline”

- Print jobs stuck in the print queue
- Unable to print test pages
- Printer not detected by workstation
- Error messages displayed during printing attempts

Possible Causes

- Printer disconnected from network
- Printer powered off
- Incorrect printer settings
- Print spooler service malfunction
- Outdated or corrupted printer drivers

Troubleshooting Steps

1. Verified that the printer was powered on and connected to the network.
2. Checked the printer display panel for error messages.
3. Restarted both the printer and workstation.
4. Cleared all stuck print jobs from the print queue.
5. Restarted the Print Spooler service through Services Manager.
6. Confirmed the printer was set as the default printer.
7. Removed and reinstalled the printer drivers.
8. Printed a test page to verify functionality.

Resolution

The issue was caused by a stuck print queue and an unresponsive Print Spooler service. Restarting the spooler service and clearing pending print jobs restored the printer to online status. Printing functionality was successfully restored.